

Sergiu-Catalin Petris

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WORK EXPERIENCE

Mentor | Codemy

Cluj-Napoca, 2025 – Present

- Taught programming basics in Scratch and LEGO Education environments (including Minecraft-based coding).
- Guided students (ages 9–12) through interactive projects and creative challenges.
- Adapted lessons to different learning speeds and encouraged problem-solving through play.

Software Developer (Java) | BMW TechWorks Romania

Cluj-Napoca, Nov 2025 – Jun 30, 2026

- Developed backend features for an enterprise Java application using Quarkus, REST APIs, Hibernate ORM/Panache, and Maven.
- Implemented and updated CRUD functionalities, DTOs, service-layer logic, validation rules, custom exceptions, and OpenAPI/Swagger documentation.
- Worked with layered backend architecture, following clean code, OOP principles, and separation of concerns between resource, service, mapper, entity, and repository classes.
- Used Git-based workflows, pull requests, code reviews, and conflict resolution while collaborating with other developers.
- Gained practical experience with enterprise development practices, API design, database persistence, error handling, and integration testing.

EDUCATION

Babeş-Bolyai University, Romania

Cluj-Napoca, 2023-Expected July 2026

Faculty of Economics and Business Administration, 3rd year

Relevant Courses:

- Object-Oriented Programming, Data Structures and Algorithms
- Databases, Operating Systems, Web Development

PERSONAL PROJECTS

Essences-MCP (*Node.js, MCP, LLMs, REST, GraphQL, RDF/SPARQL*)

- Developed an AI-driven perfume review platform that translates natural language queries into structured backend operations.
- Implemented a JSON-RPC MCP-style tool registry for dynamic tool discovery, execution, and backend routing.
- Integrated REST, GraphQL, and RDF4J/SPARQL services into a single conversational interface for Semantic Web data access.

ExplainLikeI5 (*React, Python, LLM API*)

- Full-stack educational web app simplifying complex topics into 5 points.
- React frontend with playful UI + Python backend using LLM API.
- Focused on accessibility, UX design, and scalable architecture.

Dashboard for HR (*Java, JavaFX, SQLite*)

- JavaFX desktop app for managing employees, roles, and benefits.
- Implemented CRUD operations, data serialization, and custom exception handling.
- Applied OOP principles (encapsulation, inheritance, interfaces) and design patterns.
- Work in progress – currently improving functionality and refining the interface.

VOLUNTEERING

Mentor | Cartea Daliei

Remote, 2025

Taught Scratch programming to children (ages 9–11). Designed interactive lessons to encourage creativity.

SKILLS

Programming Languages: Python, Java, C/C++, SQL, HTML/CSS/JavaScript

Frameworks: Flask, React, Tailwind, Bootstrap

Tools: Git, Linux basics

Soft Skills: Problem solving, fast learner, teamwork, communication